Midterm

February 24, 2026 10:30 AM

MATH 3042- Midterm test Full Name:
Time: 120 minutes

1. A customer support administrator monitors the number of complaints per day over 500 days. A numeric and visual summary data is as follows:

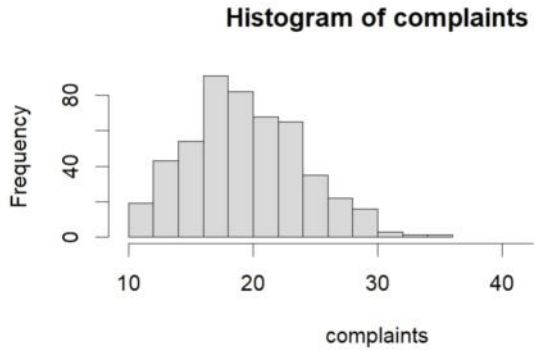
summary(complaints)

Min	Q1	Med	Mean	Q3	Max
10	17.0	19	20	23.0	36

length(complaints): 500

sd(complaints): 4.58

hist(complaints)



Based on the information above:

a. (3 marks) By calculating the upper and lower fence[1] determine if there are outliers among the Response data.

$IQR = 23 - 17 = 6$
$UF = 23 + 9 = 32$
$LF = 17 - 9 = 8$

There are upper outliers since Max =36 exceeds the UF=32
There are no lower outliers since Min =10 is greater than LF =8

b. (2 mark) Determine the z-score of Q1

$$z = \frac{17 - 20}{4.58} = -0.655$$

c. (1 mark) Does the distribution appear to be symmetric, right skewed, or left skewed?

The distribution is right-skewed (from histogram and from the fact the mean is higher than the median

d. (1 mark) The height of one of the columns on the frequency histogram of 63. What is the percentage frequency in that column?

$$percentage = \frac{63}{500} = 12.6\%$$

- [1] Lower fence = Q1 - 1.5×IQR
- Upper fence = Q3 + 1.5×IQR
- IQR = Q3-Q1
- $z = \frac{x - \bar{x}}{s}$

2. (4 marks) A microservices architecture can fail at the authentication layer (A), the processing layer (P), or the data layer (D).

Given that:

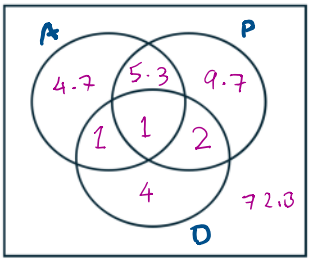
P(A) = 0.12, P(P) = 0.18, P(D) = 0.08
P(A | P) = 0.35, P(A ∩ D) = 0.02, P(P ∩ D) = 0.03, P(A ∩ P ∩ D) = 0.01

$$P(A|P) = \frac{P(A \cap P)}{P(P)}, P(A \cap P) = 0.12 \times 0.35 = 0.063$$

Determine the P(system operates successfully).

Note: Instead of P(A ∩ P) you have been given P(A | P). Use it to find P(A ∩ P).

$P(\bar{A} \cap \bar{P} \cap \bar{D}) = P(\overline{A \cup P \cup D}) = 72.3\%$



3. (3 marks) A caching system randomly maps 30 session tokens into 250 cache slots. What is the probability of at least one cache collision (two tokens assigned to the same slot)?

$$P(at\ least\ one\ collision) = 1 - P(no\ collision) = 1 - \frac{P(250,30)}{250^{30}} = 1 - 0.1632 = 0.8368$$

4. (8 marks) A distributed computing cluster runs 90 automated health checks each night. Each health check has a 96% probability of completing without errors.

a. (2 marks) What is the probability that exactly 84 health checks succeed?

$P(x = 84) = dbinom(84, 90, 0.96) = 8.27\%$

b. (2 marks) What is the probability that at least 87 health checks succeed?

$P(x \geq 87) = 1 - P(x \leq 86) = 1 - pbinom(86, 90, 0.96) = 1 - 0.4873839 \approx 51.26\%$

c. (2 marks) Calculate the expected number of successful health checks and the standard deviation.

$\mu = np, \qquad \sigma^2 = npq$

$\mu = 90 \times 0.96 = 86.4 \qquad \sigma = \sqrt{90 \times 0.96 \times 0.04} \approx 1.86$

d. (2 marks) Identify all the unusually low numbers of successful health checks. Use the $Z < -2$ criterion.

$-2 = \frac{x-86.4}{1.86}$, $x = 82.7$

Any successful health checks less than or equal to 82 is unusually low

Please use as needed:

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pbinom(84, 90, 0.96) = 0.1518973
dbinom(84, 90, 0.96) = 0.08267284
pbinom(87, 90, 0.96) = 0.7030298
pbinom(86, 90, 0.96) = 0.4873839
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5. (3 marks) A software repository contains 50 packages, of which 18 have known dependency conflicts. A developer randomly selects 10 packages to install. Using a calculator, determine the probability that at least 1 package with a dependency conflict is selected. Please write down what you are calculating.

$$P(x \geq 1) = 1 - P(x = 0) = 1 - \frac{C(18,0) \times C(32,10)}{C(50,10)} = 1 - 0.0063 = 0.9937$$

6. (4 marks) For each scenario, identify which probability distribution is most appropriate (Binomial, Hypergeometric, Geometric, or Poisson) and explain why. No need to solve the problem, just identify the appropriate distribution.

a. (1 mark) A server receives malformed HTTP requests at an average rate of 7 per hour. You want to find the probability of exactly 10 malformed requests in the next hour.

Poisson

b. (1 mark) A development team has 120 test cases, 25 of which are known to be flaky. A QA engineer randomly selects 15 test cases to run and wants to know the probability that exactly 4 are flaky.

Hypergeometric

c. (1 mark) A network router has a 12% chance of misrouting a packet on any given transmission. You want to find the probability that the first misrouted packet occurs on exactly the 6th transmission.

Geometric

d. (1 mark) A containerized application runs 60 independent startup scripts, each with a 99% chance of completing without error. You want the probability that exactly 58 scripts complete successfully.

Binomial

7. The time (in minutes) that a containerized service takes to initialize is *exponentially* [2]distributed, with a mean of 8 minutes.

A. (2 marks) What is the probability that the service initializes in under 4 minutes?

$P(x < 4) = 1 - e^{-\frac{4}{8}} = 39.3\%$

B. (2 marks) What is the probability that initialization takes between 5 and 10 minutes?

$$P(5 < x < 10) = \left(1 - e^{-\frac{10}{8}}\right) - \left(1 - e^{-\frac{5}{8}}\right) = 24.88\%$$

C. (2 marks) What is the 90th percentile of initialization times?

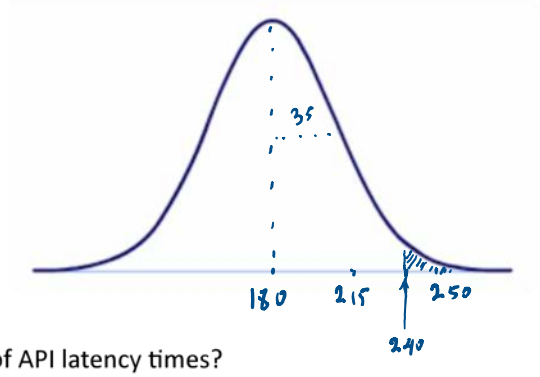
$1 - e^{-\frac{t}{8}} = 0.90$, $t = -8 \ln 0.10$, $t = 18.4$ minutes

[2] $cdf(t) = 1 - e^{-\frac{t}{\beta}}$

8. The latency (in milliseconds) of a cloud storage API under normal conditions is normally distributed, with a mean of 180 ms and a standard deviation of 35 ms.

a. (3 marks) What is the probability that a randomly chosen API call has a latency greater than 240 ms? Please place the relevant numbers on a normal curve diagram, shade the region that corresponds to the probability in question, and write the filled-in R command(s) that would solve this problem.

$P(x > 240) = 1 - pnorm(240, 180, 35)$



b. (3 marks) What values mark the middle 90% of API latency times?

Given:

$$\pm 1.64485 = \frac{x - 180}{35}, x_{low} = 122.4ms \ , x_{high} = 237.6ms$$

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> qnorm(0.05)
[1] -1.644854
> qnorm(0.10)
[1] -1.281552
```

9. A quality control team finds that 38% of software deployments require emergency patches within 24 hours. Multiple samples are taken

a. (2 marks) Calculate the mean and standard deviation of the sampling distribution of proportions for a random sample of 180 deployments requiring emergency patches.

$$\mu_{\hat{p}} = p, \quad SE = \sqrt{\frac{p(1-p)}{n}}$$

Since $(180)(0.38) and\ (180)(1-0.38)$ are both greater than or equal to 10
We can use CLT:

$\mu_{\hat{p}} = 0.38\ , SE = 0.036$

b. (2 marks) Using the normal approximation, calculate the probability that a in random sample of 180 deployments more than 43% would need emergency patches. Please write filled-in R command(s) for this part.

$P(\hat{p} > 0.43) = 1 - pnorm(0.43, 0.38, 0.036)$

c. (2 marks) At a 5% significance level, is it unusual to observe 58% patch rate in a sample of 180 deployments? Please use qnorm(0.95) = 1.644854 in your reasoning.

Let's find the z-score of the given p-hat
$z = \frac{0.56-0.38}{0.036} = 5.53$ This exceeds 1.64485 by a wide margin.
Yes, it is very unusual to observe a sample proportion this high.